Lesson plan · for the teacher

Power: batteries and the sun

IoT · Level 3 · Lesson 3

Time
40 min

Materials
A sensor card per group: battery, radio and how often it sends; Calculators or phones; Optional: an old solar garden light to open, or a small 6 V panel and a multimeter

Learning goals

- Estimate battery life from capacity and average current.
- Explain why sleeping and sending less often make a sensor last longer.
- Size a small solar panel and battery for a sensor that runs all year.

Activity: Battery-life sums

1. Warm-up (5 min): Tell the story: a soil sensor sending over Wi-Fi every minute on two AA batteries died after under three weeks of an eight-week holiday. Ask: where did the power go?
2. Teach (10 min): Write hours = capacity (mAh) ÷ average current (mA). Show the sleep, measure and send currents, and work the example: Wi-Fi once a minute averages about 4 mA, and 1,750 ÷ 4 ≈ 440 hours, about 18 days.
3. Sums (12 min): Each group gets a card with a battery, a radio and how often it sends. They work out the average current (0.01 mA for sleep, plus the send's mA × seconds ÷ the seconds between sends) and the days it lasts on 70 % of the label.
4. Challenge (8 min): Groups change one thing (how often, the radio or the battery) so their sensor lasts a term of about 90 days, and say what they changed and why. Hint: Wi-Fi every 5 minutes lasts about 90 days.
5. Wrap-up (5 min): Together, size the solar version of the Wi-Fi sensor that reports once a minute: about 100 mAh a day, 500 mAh for five cloudy days (about 715 with the 70 % rule: 500 ÷ 0.7), and a panel making about 300 mAh a day.

Prepare one sensor card per group and work the answers beforehand; a coin cell with Wi-Fi or LoRa makes a good trap card. Use batteries only, never mains. Never cut, crush or puncture lithium batteries; an adult can open an old solar garden light with a screwdriver to show the panel, battery and LED.

Discussion

- Which school sensor needs fresh data every minute, and which could report once an hour?
- Where at school would a solar panel get full sun all year?
- Who will clean the panel and check the battery each term?

Exit ticket

- A sensor averages 2 mA on 2,500 mAh of batteries. Planning with 70 %, about how many days will it last?

Try it at school tomorrow
Battery-life sums: Make a sensor card for each group: a battery (2 AA at 2,500 mAh, or a coin cell at 220 mAh), a radio (Wi-Fi: 120 mA for 2 s; LoRa: 40 mA for 0.4 s; Bluetooth LE: 10 mA for 0.05 s) and how often it sends. A coin cell gives only about 20 mA at a time, so it can run only the Bluetooth LE card: a good trap to spot. Each group works out the average current and how many days it lasts with the 70 % rule, then changes one thing to make it last a whole term of about 90 days. Finish with one solar sum on the board: a battery for five cloudy days and a panel three times the daily use. If you have an old solar garden light, open it to show the panel, battery and LED.

Practice questions · for you

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1 **Amani's garden sensor has a battery that covers five cloudy days, but its solar panel only just covers one day's use in full sun. What happens over the year?**

- ◯ It runs all year: the panel covers each day's use
- ◯ It runs all year, as long as the battery is bigger
- ◯ It stops after cloudy spells: the battery never refills

2 **A soil sensor sends over Bluetooth LE every 30 seconds and averages about 0.05 mA. Its two AA batteries hold 2,500 mAh.**

Planning with 70 %, it has _____ mAh to use. Divided by 0.05 mA, that's _____ hours: about four years.

Word bank: 35,000 · 50,000 · 1,750 · 2,500

3 **A coin cell (CR2032) is a good battery for a Wi-Fi sensor, as long as it reports only once an hour.**

- ◯ True
- ◯ False

4 **In the battery lab, with two AA batteries (the lab uses the full 2,500 mAh, with no 70 % margin), which of these are true?**

- ☐ Wi-Fi every hour lasts over three years
- ☐ Reporting every 6 hours, a power bank outlasts two AAs
- ☐ Bluetooth LE every minute lasts years
- ☐ Wi-Fi every minute lasts under a month
- ☐ A LoRa send costs more than a Wi-Fi send

5 **Put the steps for sizing a solar-powered sensor in order.**

- ☐ Divide by 0.7 to choose the battery
- ☐ Choose a panel that makes about 3 times the daily use
- ☐ Multiply by 24 for the daily use (mAh)
- ☐ Work out the sensor's average current (mA)
- ☐ Multiply by the cloudy days to cover

✎ **Which sensor at your school would you power with the sun, and how fresh does its data really need to be?**

Check your answers

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1 **It stops after cloudy spells: the battery never refills**

The panel has nothing spare, so every cloudy day takes charge that sunny days can't give back.

2 **Planning with 70 %, it has 1,750 mAh to use. Divided by 0.05 mA, that's 35,000 hours: about four years.**

70 % of 2,500 is 1,750 mAh, and 1,750 ÷ 0.05 = 35,000 hours; divided by 24, that's about 1,460 days, or four years.

3 **False**

A coin cell can only give about 20 mA at a time, and a Wi-Fi send needs about 120 mA, so the sensor would keep restarting however rarely it reports.

4 **Wi-Fi every minute lasts under a month; Wi-Fi every hour lasts over three years; Bluetooth LE every minute lasts years**

Wi-Fi every minute averages about 4 mA: weeks, not months.

5 **Top to bottom: 4, 5, 2, 1, 3**

Everything starts from the average current.